FINAL YEAR ENGINEERING PROJECT

ORBITAL TIG WELDING FOR INDUSTRIAL PIPING

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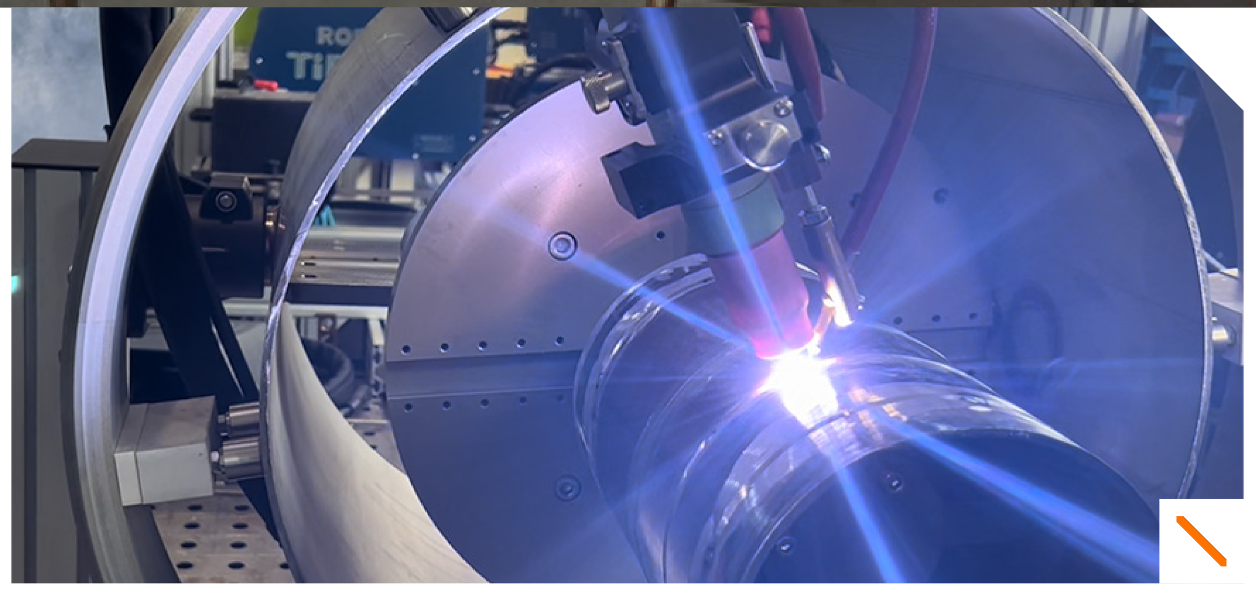
THE COMPANY

- **CIMAT SARTEC**, FOSELEV Group. Industrial piping and maintenance.
- Chemical, petrochemical, manufacturing and steel industries.
- Boilermaking, steel structures, piping and pressure equipment.
- 1155 boulevard du Grand Terme, 30133 Les Angles.



ORBITAL TIG WELDING

- Torch carried right around a pipe held in **fixed position**.
- **Pulsed current**: the molten pool stays under control.
- **Multiple angular sectors**, one set of parameters for each welding position.
- Enclosed head, argon shielding and **back purging** of the root.

WHAT WOULD CHANGE

- **Today**: every pipe weld made by hand, the result tied to the welder.
- **Tomorrow**: programmed travel, a bead that comes out the same every time.
- Roughly **half the welding time** on the joint taken as a reference.
- Real welding parameters **logged automatically**, weld after weld.

WORTH THE INVESTMENT

Welding times to be confirmed by measurement in the workshop before the order is signed.

WHAT COMES NEXT

- Welding on site, alongside the teams.
- Open head fed with wire for large diameters and thick walls.
- A library of welding programmes that gains value over time.

MY ROLE

Apprentice project engineer: estimating, supplier consultation, planning, fabrication follow-up and inspection, through to installation on site.

SCOPE STUDIED

Austenitic stainless steel 304L and 316L, small and medium diameters, the everyday production of the workshop.

COMPLIANCE PATH

Welding procedure qualified first, then the operators to ISO 14732, then the usual inspections. Nothing stands in the way.

Finished orbital weld on 316L: 4X4 Blazer 1776, Wikimedia Commons, CC BY 2.0. Other photos: FOSELEV and equipment manufacturers.